12.775

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Question

Choose solution set S corresponding to the systems of two equations

$$x - 2y + z = 0$$
$$x - z = 0$$

1. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\}$
2. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$
3. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$
4. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\}$

Theoretical Solution

We are given a homogeneous system of two linear equations with three variables (x, y, z). We need to find all vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ that satisfy both equations simultaneously.
We can solve this system using substitution.

Theoretical Solution

- **From the second equation:**

$$x - z = 0 \implies x = z \tag{1}$$

- **Substitute $x = z$ into the first equation:**

$$\begin{aligned} (z) - 2y + z &= 0 \quad (2)\\ 2z - 2y &= 0 \quad (3)\\ 2z &= 2y \quad (4)\\ z &= y \quad (5) \end{aligned}$$

- **Combine the results:** We have found that $x = z$ and $y = z$. Therefore, all three variables must be equal:

$$x = y = z \tag{6}$$

Theoretical Solution

Since we have three variables (x, y, z) but only two independent equations, we have one free variable. We can express the solution set in terms of a single parameter, let's call it $\alpha \in \mathbb{R}$.

- Let the free variable be $z = \alpha$.
- Since $y = z$, then $y = \alpha$.
- Since $x = z$, then $x = \alpha$.

The solution vectors are of the form:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha \\ \alpha \\ \alpha \end{pmatrix} \tag{7}$$

Theoretical Solution

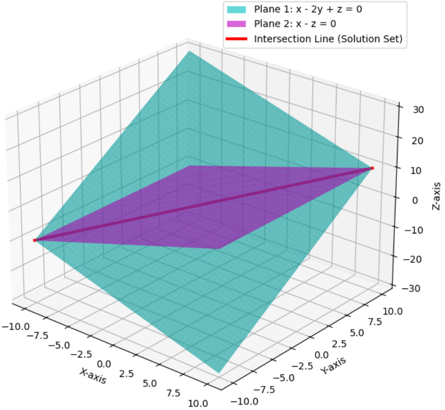
We can factor out the parameter α:

$$\alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{8}$$

The solution set S is the set of all possible vectors of this form for any real number α.
Therefore, the solution set is:

$$S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\} \tag{9}$$

Plot


Plane 1: x - 2y + z = 0
Plane 2: x - z = 0
Intersection Line (Solution Set)
X-axis
Y-axis
Z-axis
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
30
20
10
0
−10
−20
−30